

Periodogram and Discrete Fourier Transform

STAT 429: Introduction to Time Series
Week 4

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Contents

1	Efficient Computation of the Residual Sum of Squares	2
1.1	Formula for $RSS(f)$ when f is a Fourier Frequency	2
1.2	Proof of the orthogonality identities	4
1.3	The Periodogram	4
2	Discrete Fourier Transform and Periodogram	5
2.1	Basic Properties of the DFT	6
3	Complex Sinusoids and Orthogonality	6
3.1	Complex Sinusoids	6
3.2	Complex Sinusoidal Vectors	7
3.3	The Inverse DFT formula	8
4	Frequency Inference on the Fourier Grid	9

1 Efficient Computation of the Residual Sum of Squares

Continuing the sinusoidal model from Week 3, write

$$y_t = \beta_0 + \beta_1 \cos(2\pi ft) + \beta_2 \sin(2\pi ft) + \epsilon_t, \quad \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2). \quad (1)$$

Here $y = (y_1, \dots, y_n)^T$, $\beta = (\beta_0, \beta_1, \beta_2)^T$, and row t of X_f is $(1, \cos(2\pi ft), \sin(2\pi ft))$. The quantity

$$RSS(f) = \min_{\beta} \|y - X_f \beta\|^2$$

measures how well frequency f fits the data. It is used both to estimate f and to evaluate its posterior from Week 3. A plot of $RSS(f)$ is also an exploratory tool for identifying periodicities, even when we do not intend to fit the single-frequency model.

For computing $RSS(f)$, we need a grid of values for f . The most commonly used grid is given by:

$$\mathcal{F} := \left\{ 0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{[n/2]}{n} \right\} \quad (2)$$

where $[n/2]$ is the largest integer smaller than or equal to $n/2$.

A frequency of the form j/n where $j \in \{0, 1, 2, \dots, n-1\}$ and n is the observed data size is called a **Fourier Frequency**. So the grid (2) consists of all Fourier frequencies that are in the range $[0, 1/2]$.

The main reason for taking the grid to consist of Fourier Frequencies is that $RSS(f)$, $f \in \mathcal{F}$ can be computed very efficiently (in time $O(n \log n)$) using a classical algorithm known as the Fast Fourier Transform (FFT). We explain the high-level details behind this fact (without going into the workings of the FFT algorithm).

1.1 Formula for $RSS(f)$ when f is a Fourier Frequency

When f is a Fourier Frequency, one can write down a more explicit formula for $RSS(f)$. For this, first note that whenever X_f has full column rank:

$$RSS(f) = \min_{\beta} \|y - X_f \beta\|^2 = \|y - X_f \hat{\beta}_f\|^2 \quad \text{where } \hat{\beta}_f := (X_f^T X_f)^{-1} X_f^T y.$$

Expanding the square and using $X_f^T X_f \hat{\beta}_f = X_f^T y$ gives

$$\begin{aligned} RSS(f) &= y^T y - 2 \hat{\beta}_f^T X_f^T y + \hat{\beta}_f^T X_f^T X_f \hat{\beta}_f \\ &= y^T y - y^T X_f (X_f^T X_f)^{-1} X_f^T y. \end{aligned}$$

Now

$$X_f^T X_f = \begin{pmatrix} n & \sum_{t=1}^n \cos(2\pi ft) & \sum_{t=1}^n \sin(2\pi ft) \\ \sum_{t=1}^n \cos(2\pi ft) & \sum_{t=1}^n \cos^2(2\pi ft) & \sum_{t=1}^n \cos(2\pi ft) \sin(2\pi ft) \\ \sum_{t=1}^n \sin(2\pi ft) & \sum_{t=1}^n \cos(2\pi ft) \sin(2\pi ft) & \sum_{t=1}^n \sin^2(2\pi ft) \end{pmatrix}$$

Now suppose that $f \in (0, 0.5)$ and suppose that f is a Fourier frequency i.e., it is of the form $f = j/n$ for some integer j . Then it turns out that

$$\begin{aligned} \sum_{t=1}^n \cos(2\pi ft) &= 0 & \sum_{t=1}^n \sin(2\pi ft) &= 0 \\ \sum_{t=1}^n \cos^2(2\pi ft) &= \frac{n}{2} & \sum_{t=1}^n \sin^2(2\pi ft) &= \frac{n}{2} \\ \sum_{t=1}^n \cos(2\pi ft) \sin(2\pi ft) &= 0 \end{aligned} \quad (3)$$

As a result, for such f ,

$$X_f^T X_f = \begin{pmatrix} n & 0 & 0 \\ 0 & n/2 & 0 \\ 0 & 0 & n/2 \end{pmatrix} \quad \text{so that} \quad (X_f^T X_f)^{-1} = \begin{pmatrix} 1/n & 0 & 0 \\ 0 & 2/n & 0 \\ 0 & 0 & 2/n \end{pmatrix}$$

Since

$$X_f^T y = \begin{pmatrix} \sum_t y_t \\ \sum_t y_t \cos(2\pi f t) \\ \sum_t y_t \sin(2\pi f t) \end{pmatrix},$$

we obtain

$$\begin{aligned} y^T X_f (X_f^T X_f)^{-1} X_f^T y &= n\bar{y}^2 + \frac{2}{n} \left(\sum_t y_t \cos(2\pi f t) \right)^2 \\ &\quad + \frac{2}{n} \left(\sum_t y_t \sin(2\pi f t) \right)^2. \end{aligned}$$

Therefore for Fourier frequencies in the range $(0, 0.5)$, we get

$$RSS(f) = y^T y - n\bar{y}^2 - \frac{2}{n} \left(\sum_t y_t \cos(2\pi f t) \right)^2 - \frac{2}{n} \left(\sum_t y_t \sin(2\pi f t) \right)^2$$

or equivalently

$$RSS(f) = \sum_t (y_t - \bar{y})^2 - \frac{2}{n} \left(\sum_t y_t \cos(2\pi f t) \right)^2 - \frac{2}{n} \left(\sum_t y_t \sin(2\pi f t) \right)^2 \quad (4)$$

When $f = 0$ and $f = 1/2$, the above formula needs to be slightly modified. When $f = 0$, the sinusoidal model (1) simply becomes:

$$y_t = \beta_0 + \beta_1 + \epsilon_t \quad \text{with } \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

There is only one effective coefficient parameter $(\beta_0 + \beta_1)$ here which will be estimated by $\bar{y}$ so that RSS becomes

$$RSS(0) = \sum_t (y_t - \bar{y})^2. \quad (5)$$

When $f = 1/2$ and n is even (when n is odd, $1/2$ cannot be a Fourier frequency so it will not be considered), the model (1) becomes

$$y_t = \beta_0 + \beta_1 \cos(\pi t) + \epsilon_t = \beta_0 + \beta_1 (-1)^t + \epsilon_t.$$

For this model, it is easy to check that

$$X^T X = \begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix}$$

so that

$$RSS(1/2) = \sum_t (y_t - \bar{y})^2 - \frac{1}{n} \left(\sum_t y_t (-1)^t \right)^2 \quad (6)$$

Let us not worry too much about the edge cases $f = 0$ and $f = 1/2$, and focus on the formula (4). Note again that this formula holds whenever $f \in (0, 0.5)$ and f is a Fourier frequency (i.e., nf is an integer).

1.2 Proof of the orthogonality identities

Suppose $0 < f < 1/2$ and nf is an integer. The geometric-series formula gives

$$\sum_{t=1}^n e^{2\pi i f t} = \frac{e^{2\pi i f} (e^{2\pi i n f} - 1)}{e^{2\pi i f} - 1} = 0.$$

The denominator is nonzero because $0 < f < 1/2$, and the numerator vanishes because nf is an integer. Taking real and imaginary parts gives

$$\sum_{t=1}^n \cos(2\pi f t) = 0, \quad \sum_{t=1}^n \sin(2\pi f t) = 0.$$

Similarly,

$$\sum_{t=1}^n e^{4\pi i f t} = \frac{e^{4\pi i f} (e^{4\pi i n f} - 1)}{e^{4\pi i f} - 1} = 0,$$

since $2nf$ is an integer and $0 < 2f < 1$. Hence the sums of $\cos(4\pi f t)$ and $\sin(4\pi f t)$ both vanish. Using

$$\begin{aligned} \cos^2(2\pi f t) &= \frac{1 + \cos(4\pi f t)}{2}, \\ \sin^2(2\pi f t) &= \frac{1 - \cos(4\pi f t)}{2}, \\ \cos(2\pi f t) \sin(2\pi f t) &= \frac{1}{2} \sin(4\pi f t), \end{aligned}$$

proves the remaining identities in (3).

1.3 The Periodogram

Given a time series dataset $y_1, \dots, y_n$, its periodogram is the function $I(f)$, $0 < f < 1/2$, defined as follows:

$$I(f) := \frac{1}{n} \left(\sum_{t=1}^n y_t \cos(2\pi f t) \right)^2 + \frac{1}{n} \left(\sum_{t=1}^n y_t \sin(2\pi f t) \right)^2 \quad \text{for } f \in (0, 0.5)$$

From the formula (4), we have

$$RSS(f) = \sum_t (y_t - \bar{y})^2 - 2I(f) \quad \text{when } f \in (0, 0.5) \text{ is a Fourier Frequency.}$$

The periodogram $I(f)$ can be written in the following alternative way:

$$I(f) = \frac{1}{n} \left| \sum_{t=1}^n y_t e^{-2\pi i f t} \right|^2$$

where $|\cdot|$ denotes complex modulus. At Fourier frequencies the sum inside the modulus is closely related to the discrete Fourier transform, which can be computed efficiently by the FFT.

2 Discrete Fourier Transform and Periodogram

The sum in the complex-modulus expression for the periodogram is almost the DFT; the difference is an indexing convention. While discussing the DFT, it is a standard convention to write the data as $y_0, y_1, \dots, y_{n-1}$ (instead of $y_1, \dots, y_n$). In other words, we start the indexing with 0 (as in Python) as opposed to 1. With this indexing, the DFT is defined as follows.

The DFT of data $y_0, \dots, y_{n-1}$ is defined by:

$$b_j := \sum_{t=0}^{n-1} y_t \exp\left(-\frac{2\pi i j t}{n}\right) \quad \text{for } j = 0, 1, \dots, n-1. \quad (7)$$

In words, b_j is the complex inner product between the data and the complex sinusoid with frequency $f = j/n$. Note that the complex sinusoid with frequency f is given by:

$$\exp(2\pi i f t) \quad \text{for } t = 0, 1, 2, \dots, n-1.$$

The coefficients $b_0, \dots, b_{n-1}$ collectively form the DFT. Even for real-valued data, these coefficients can be complex-valued.

More on indexing: As mentioned previously, while defining the DFT, we index the data starting from zero. In the definition (7), the first data point (which is y_0) is being multiplied by $\exp(-2\pi i j(0)/n) = 1$, the second data point is being multiplied by $\exp(-2\pi i j/n)$ and, in general, the t -th data point is being multiplied by $\exp(-2\pi i j(t-1)/n)$. If we instead define the DFT by:

$$\tilde{b}_j := \sum_{t=1}^n y_t \exp\left(-\frac{2\pi i j t}{n}\right),$$

then the t -th data point will be multiplied by $\exp(-2\pi i j t/n)$. b_j and $\tilde{b}_j$ will be different and they will be related by:

$$b_j = \tilde{b}_j \exp\left(\frac{2\pi i j}{n}\right).$$

The multiplier $\exp(2\pi i j/n)$ above has modulus one which means that

$$|b_j| = |\tilde{b}_j|.$$

So if we are only looking at the moduli of the DFT terms (note the periodogram only involves the moduli of DFT), then it does not matter whether we index data starting with 0 or 1. The standard convention while studying the DFT is to index starting from 0.

The connection between the periodogram and the DFT is given by:

$$I(j/n) = \frac{|b_j|^2}{n} \quad \text{for } 0 < \frac{j}{n} < 1/2$$

It is important to note that the DFT can be calculated very efficiently. The DFT of $y = (y_0, \dots, y_{n-1})^T$ can be obtained in `numpy` using the command `np.fft.fft(y)`. Here `fft` stands for Fast Fourier Transform which is a special efficient algorithm for computing the DFT. Naively, it would seem that to compute the DFT would need $O(n^2)$ computation (for each of n values of j , we have to compute b_j which requires a sum over the dataset of size n), but the FFT algorithm exploits symmetry and redundancy in the complex exponentials to compute the DFT in only $O(n \log n)$ time. We will not be going over the details of the FFT algorithm in class.

2.1 Basic Properties of the DFT

Here are some basic things to note about the DFT (defined in (7)).

1. b_0 is always equal to $y_0 + \dots + y_{n-1}$. To see this, just plug in $j = 0$ in (7).
2. In general b_j is a complex number with real and imaginary parts given by:

$$\text{real part of } b_j = \sum_{t=0}^{n-1} y_t \cos\left(\frac{2\pi jt}{n}\right)$$

$$\text{imaginary part of } b_j = -\sum_{t=0}^{n-1} y_t \sin\left(\frac{2\pi jt}{n}\right)$$

Sometimes the imaginary part will be zero (for example, when n is even and $j = n/2$) but, in general, b_j will be complex-valued.

3. For each $j = 1, \dots, n-1$, the DFT term b_{n-j} equals the complex conjugate of b_j :

$$b_{n-j} = \bar{b}_j. \quad (8)$$

The reason for the above is

$$b_{n-j} = \sum_t y_t \exp\left(-\frac{2\pi i(n-j)t}{n}\right) = \sum_t y_t \exp\left(\frac{2\pi ijt}{n}\right) \exp(-2\pi it) = \bar{b}_j,$$

where, in the above, we used that $\exp(-2\pi it) = 1$ (because t is an integer) and that $\exp\left(\frac{2\pi ijt}{n}\right)$ is the complex conjugate of $\exp\left(-\frac{2\pi ijt}{n}\right)$. Note that, for the above argument, it is crucial that $y_0, \dots, y_{n-1}$ are real. If some of $y_0, \dots, y_{n-1}$ are complex, the relation (8) is no longer true.

Because of (8), the DFT terms for later indices j can be determined as complex conjugates for the DFT terms for earlier indices. For example, when $n = 11$, the DFT can be written as:

$$b_0, b_1, b_2, b_3, b_4, b_5, \bar{b}_5, \bar{b}_4, \bar{b}_3, \bar{b}_2, \bar{b}_1,$$

and, for $n = 12$, it is

$$b_0, b_1, b_2, b_3, b_4, b_5, b_6 = \bar{b}_6, \bar{b}_5, \bar{b}_4, \bar{b}_3, \bar{b}_2, \bar{b}_1.$$

Note that when $n = 12$, the term b_6 is necessarily real because $b_6 = \bar{b}_6$.

Thus when $n = 11$, the data consists of 11 real numbers while the DFT consists of one real number (b_0) and 5 complex numbers. On the other hand, when $n = 12$, the data consists of 12 real numbers while the DFT consists of two real numbers (b_0 and b_6) and 5 complex numbers.

It turns out that the data $y_0, y_1, \dots, y_{n-1}$ can be recovered from the DFT $b_0, \dots, b_{n-1}$ using a simple formula (this formula is sometimes known as the inverse DFT formula). Before seeing this, it is necessary to understand orthogonality properties of complex sinusoids with Fourier frequencies.

3 Complex Sinusoids and Orthogonality

3.1 Complex Sinusoids

As in Week 3, sinusoids are linear combinations of $\cos(2\pi ft)$ and $\sin(2\pi ft)$. While doing algebra with sinusoids, it is very useful to represent them in terms of complex exponentials as:

$$\cos(2\pi ft) = \frac{1}{2} \exp(2\pi ift) + \frac{1}{2} \exp(-2\pi ift)$$

$$\sin(2\pi ft) = \frac{1}{2i} \exp(2\pi ift) - \frac{1}{2i} \exp(-2\pi ift)$$

In Week 3, we saw that while dealing with sinusoids at integer valued time points t , we can restrict the frequency f to $[0, 1/2]$. However, if we use the formulae above, note that we have to deal with $-f$ as well (because of the second term $e^{-2\pi ift} = e^{2\pi i(-f)t}$) and $-f$ lies between $-1/2$ and 0 . Thus when discussing sinusoids in terms of complex exponentials $e^{2\pi ift}$, $t = 0, 1, \dots, n-1$, one takes $f \in [-0.5, 0.5]$ (note that $f = -0.5$ leads to the same $e^{2\pi ift}$ as $f = 0.5$ so we drop $f = 0.5$ from consideration). For example, the function `np.fft.fftfreq(n)` gives all Fourier frequencies in $[-0.5, 0.5]$.

If one does not want to deal with negative frequencies, then we can use

$$e^{-2\pi ift} = \cos(2\pi ft) - i \sin(2\pi ft) = \cos(2\pi(1-f)t) + i \sin(2\pi(1-f)t) = e^{2\pi i(1-f)t}$$

because $\cos(2\pi(1-f)t) = \cos(2\pi t - 2\pi ft) = \cos(2\pi ft)$ (note t is an integer) and $\sin(2\pi(1-f)t) = \sin(2\pi t - 2\pi ft) = -\sin(2\pi ft)$.

Therefore, if we want to use complex exponentials $e^{2\pi ift}$ but we do not want to deal with negative frequencies, then we can restrict f to $[0, 1)$. From here on, whenever we consider the complex sinusoid $x_t = e^{2\pi ift}$ for $t = 0, 1, \dots, n-1$, we restrict $f \in [0, 1)$.

3.2 Complex Sinusoidal Vectors

For every $0 \leq j \leq (n-1)$, let us define the $n \times 1$ vector

$$u^j = (1, \exp(2\pi ij/n), \exp(2\pi i2j/n), \dots, \exp(2\pi i(n-1)j/n))^T.$$

This vector can be interpreted as the complex sinusoid $e^{2\pi ift}$ with Fourier frequency $f = j/n$ evaluated at the time points $t = 0, 1, \dots, (n-1)$. It is easy to see that

1. When $j = 0$, we have $u^0 = (1, 1, \dots, 1)$.
2. When $1 \leq j \leq n-1$, we have $u^j = \overline{u^{n-j}}$. Here $\bar{u}$ denotes complex conjugate of u (the complex conjugate $\bar{u}$ of a vector u is defined as the vector obtained by taking the complex conjugates of each entry of u).

The most important property of these complex valued vectors $u^0, u^1, \dots, u^{n-1}$ is **orthogonality**. Specifically, for $0 \leq j \neq k \leq n-1$, we have

$$\langle u^j, u^k \rangle = 0. \tag{9}$$

Recall that the inner product between two complex valued vectors $a = (a_1, \dots, a_n)^T$ and $b = (b_1, \dots, b_n)^T$ is given by

$$\langle a, b \rangle = \sum_{j=1}^n a_j \bar{b}_j.$$

Note specially the complex conjugate of b_j above.

Here is the proof of (9). Fix $0 \leq j \neq k \leq n-1$ and write

$$\begin{aligned}
\langle u^j, u^k \rangle &= \sum_{t=0}^{n-1} \exp\left(2\pi i \frac{j}{n} t\right) \overline{\exp\left(2\pi i \frac{k}{n} t\right)} \\
&= \sum_{t=0}^{n-1} \exp\left(2\pi i \frac{j}{n} t\right) \exp\left(-2\pi i \frac{k}{n} t\right) \\
&= \sum_{t=0}^{n-1} \exp\left(2\pi i \frac{j-k}{n} t\right) \\
&= \sum_{t=0}^{n-1} \left[\exp\left(2\pi i \frac{j-k}{n}\right) \right]^t \\
&= \frac{1 - \left(\exp\left(2\pi i \frac{j-k}{n}\right)\right)^n}{1 - \exp\left(2\pi i \frac{j-k}{n}\right)} \\
&= \frac{1 - \exp(2\pi i(j-k))}{1 - \exp\left(2\pi i \frac{j-k}{n}\right)} \\
&= \frac{1 - \cos(2\pi(j-k)) - i \sin(2\pi(j-k))}{1 - \exp\left(2\pi i \frac{j-k}{n}\right)} = \frac{1 - 1 - 0}{1 - \exp\left(2\pi i \frac{j-k}{n}\right)} = 0
\end{aligned}$$

This proves (9). It is also easy to see that (just take $j = k$ in the above calculation and the answer can be found in the third line)

$$\langle u^j, u^j \rangle = \|u^j\|^2 = n.$$

Therefore the n complex-valued vectors $u^0, u^1, \dots, u^{n-1}$ are orthogonal and they all have the same squared length equal to n . This immediately implies that they form a **basis** for the space $\mathbb{C}^n$ consisting of all complex-valued vectors of length n . In other words, every complex-valued vector of length n can be written as a linear combination of $u^0, u^1, \dots, u^{n-1}$.

3.3 The Inverse DFT formula

The inverse DFT formula recovers the data $y_0, \dots, y_{n-1}$ from their DFT $b_0, \dots, b_{n-1}$. We derive this formula below.

The main observation is the following: Because $u^0, \dots, u^{n-1}$ form a basis, we can write any $n \times 1$ vector of complex entries:

$$y = (y_0, \dots, y_{n-1})^T$$

as a linear combination of $u^0, \dots, u^{n-1}$. More specifically, we can write

$$y = a_0 u^0 + a_1 u^1 + \dots + a_{n-1} u^{n-1} \quad (10)$$

Take the inner product of both sides of the above equation with u^j for a fixed j and use orthogonality so that $\langle u^j, u^k \rangle = 0$ for $k \neq j$ and the fact that $\langle u^j, u^j \rangle = n$ to obtain

$$a_j = \frac{1}{n} \langle y, u^j \rangle = \frac{1}{n} \sum_{t=0}^{n-1} y_t \exp\left(-\frac{2\pi i j t}{n}\right). \quad (11)$$

By the formula (7) for the DFT b_j , it is easy to see that $a_j = b_j/n$. As a consequence (10) becomes:

$$y = \frac{1}{n} (b_0 u^0 + b_1 u^1 + \dots + b_{n-1} u^{n-1}).$$

Writing the t -th entry on both sides, we get

$$y_t = \frac{1}{n} \sum_{j=0}^{n-1} b_j \exp\left(\frac{2\pi i j t}{n}\right) \quad \text{for each } t = 0, 1, \dots, n-1. \quad (12)$$

This is the inverse DFT formula. Note that the inverse DFT formula (12) as well as the DFT definition (7) look similar; the differences being in the sign of the exponent in the complex exponential and the presence of the factor $1/n$ in (12).

4 Frequency Inference on the Fourier Grid

For numerical inference on f , we can use a dense grid in $(0, 1/2)$ or restrict attention to the Fourier frequencies in this interval. The calculations above give an efficient procedure for the latter choice:

1. Compute the DFT with `np.fft.fft(y)`.
2. Obtain $I(j/n) = |b_j|^2/n$ and then

$$RSS(j/n) = \sum_{t=0}^{n-1} (y_t - \bar{y})^2 - 2I(j/n), \quad 0 < j/n < 1/2.$$

3. Minimize these residual sums of squares to estimate f .
4. Evaluate the posterior weights on this grid using

$$\text{posterior}(j/n) \propto \left(\frac{1}{RSS(j/n)} \right)^{(n-3)/2}, \quad 0 < j/n < 1/2.$$

In the posterior expression from Week 3, there was also a factor $|X_f^T X_f|^{-1/2}$. For the interior Fourier frequencies,

$$X_f^T X_f = \text{diag}(n, n/2, n/2), \quad |X_f^T X_f| = \frac{n^3}{4}.$$

This determinant does not depend on f , so its contribution is absorbed in the proportionality constant.

Restricting to Fourier frequencies is a computational choice. It can lose information compared with a denser grid; the sunspots dataset is an example mentioned in the lectures.